



Critical System Hang in Linux UACCE Architecture

Technical analysis, exploit mechanics, and remediation for CVE-2026-23096.

TARGET_FILE: drivers/misc/uacce/uacce.c	CWE_ID: 415/672	VULN_TYPE: Double Cleanup
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TECH-SPEC

COMPONENT :

UACCE FRAMEWORK

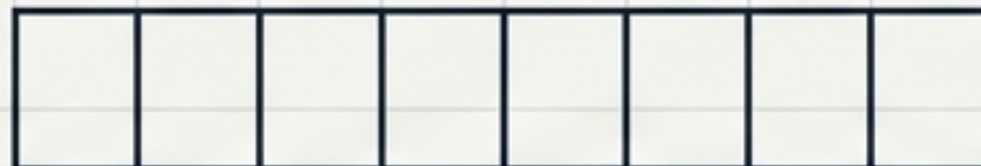
VENDOR :

LINUX KERNEL

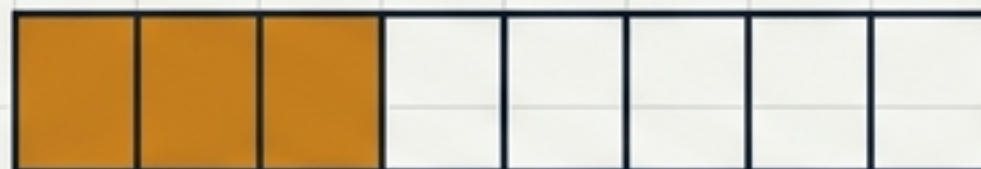
FLAW :

IMPROPER RESOURCE
MANAGEMENT

IMPACT GAUGE



CONFIDENTIALITY: NONE (0%)



INTEGRITY: LOW
(MINOR STATE CORRUPTION)

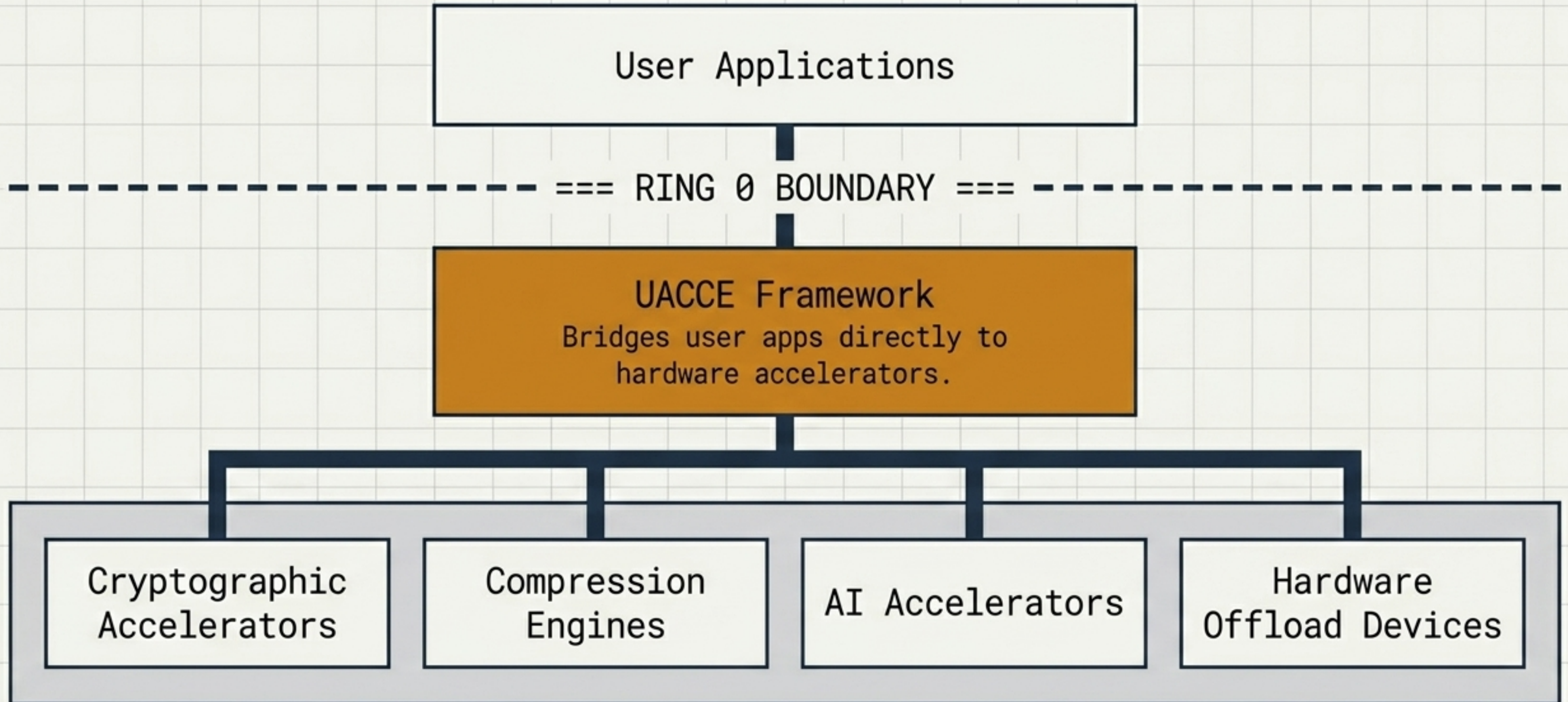


AVAILABILITY: CRITICAL
(KERNEL HANG / COMPLETE FREEZE)

ALERT

A LOGIC ERROR IN THE
CHARACTER DEVICE
CLEANUP PATH
TRIGGERS A DUPLICATE
RESOURCE REMOVAL,
PERMANENTLY
HANGING THE KERNEL
AND CAUSING
CAUSING IMMEDIATE
DENIAL OF SERVICE.

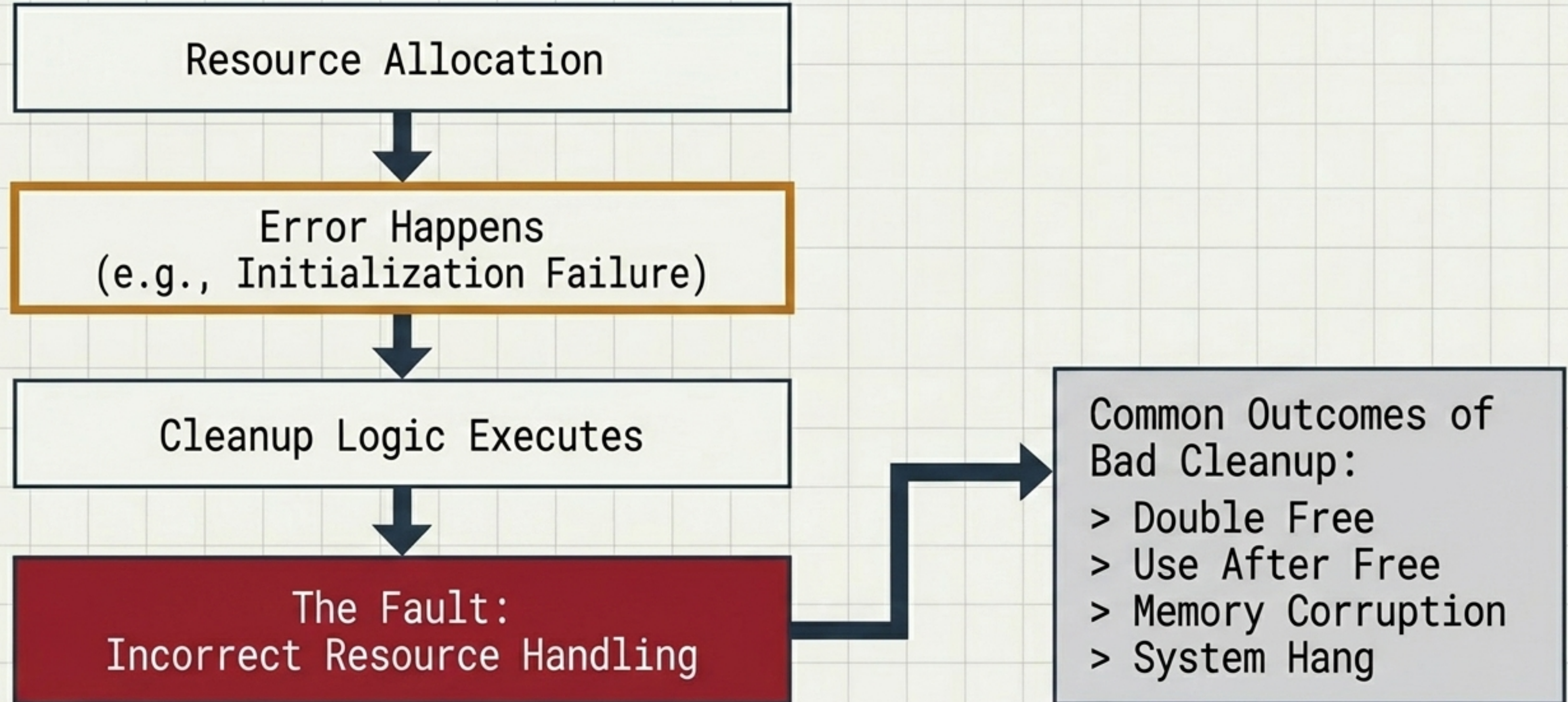
The UACCE Abstraction Layer



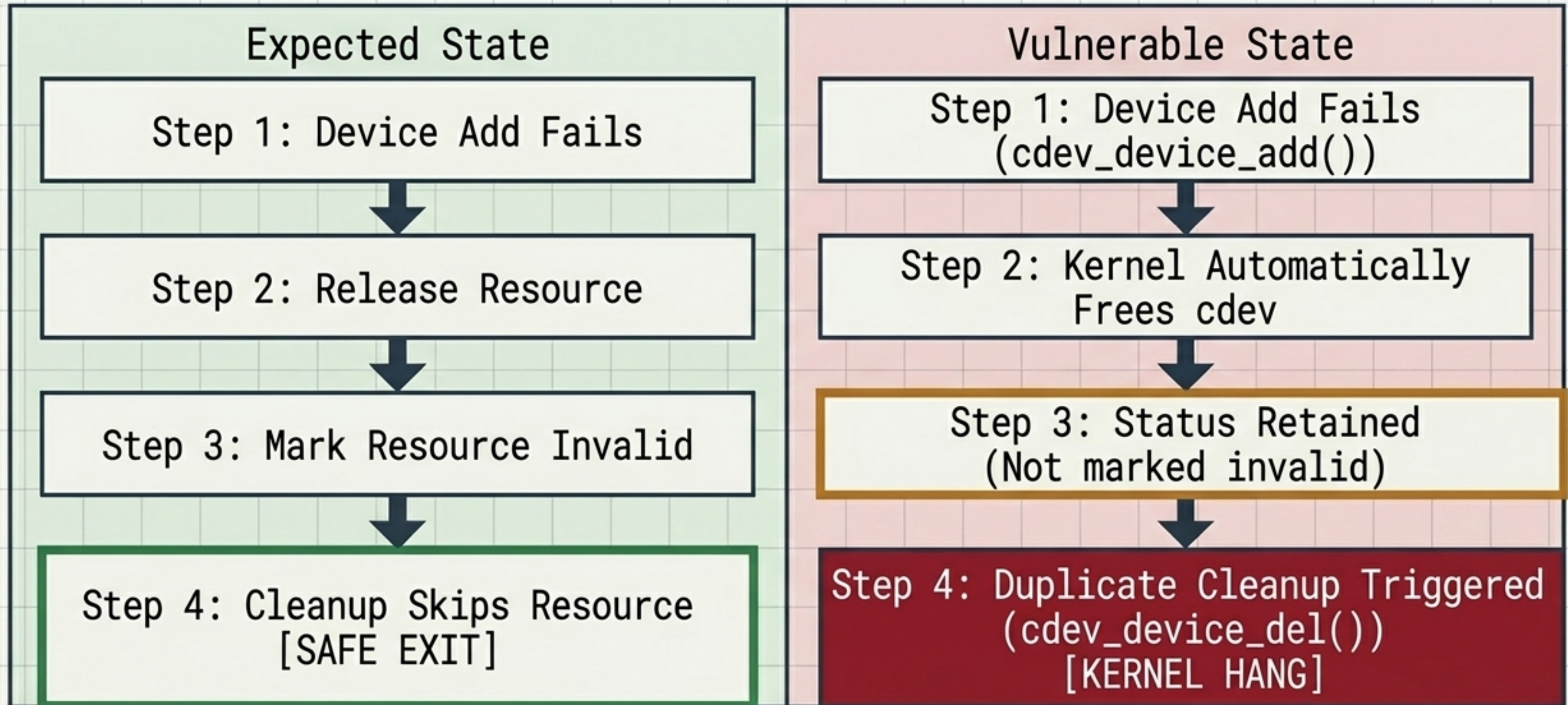
Exploitation Prerequisites

Required Conditions	Conditions Not Required
[✓] Local Access to System	[✗] Remote Network Execution
[✓] Ability to Trigger Device Initialization Failure	[✗] Secondary User Interaction
[✓] Vulnerable Kernel Version	[✗] Inbound Network Connectivity

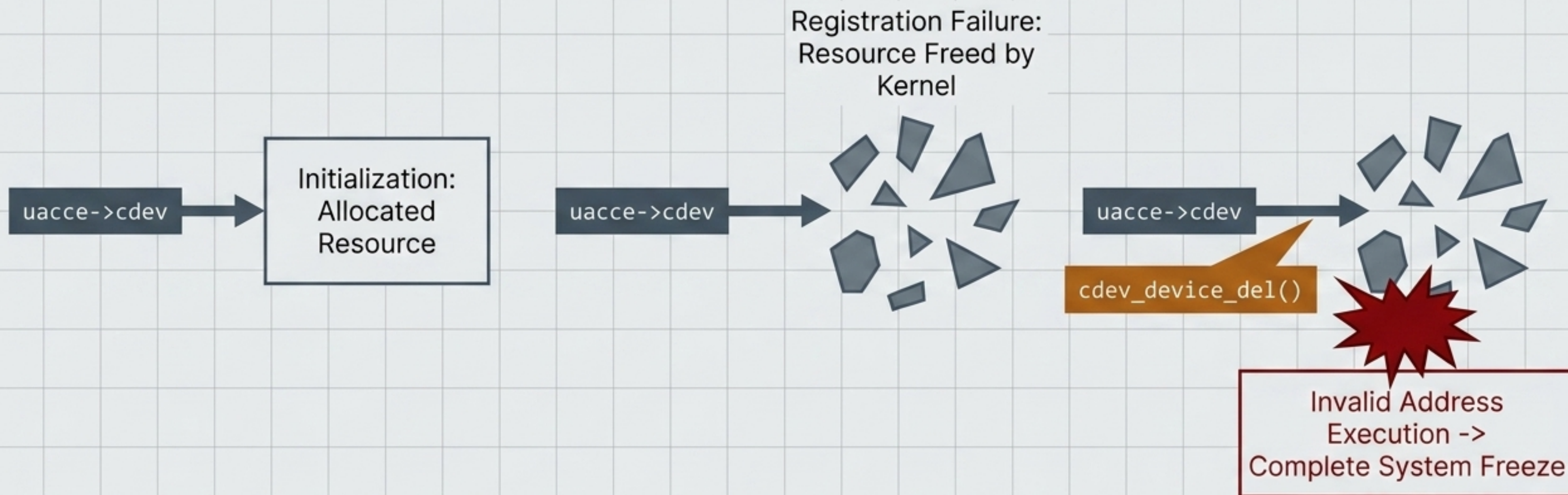
Anatomy of a Cleanup Path Vulnerability



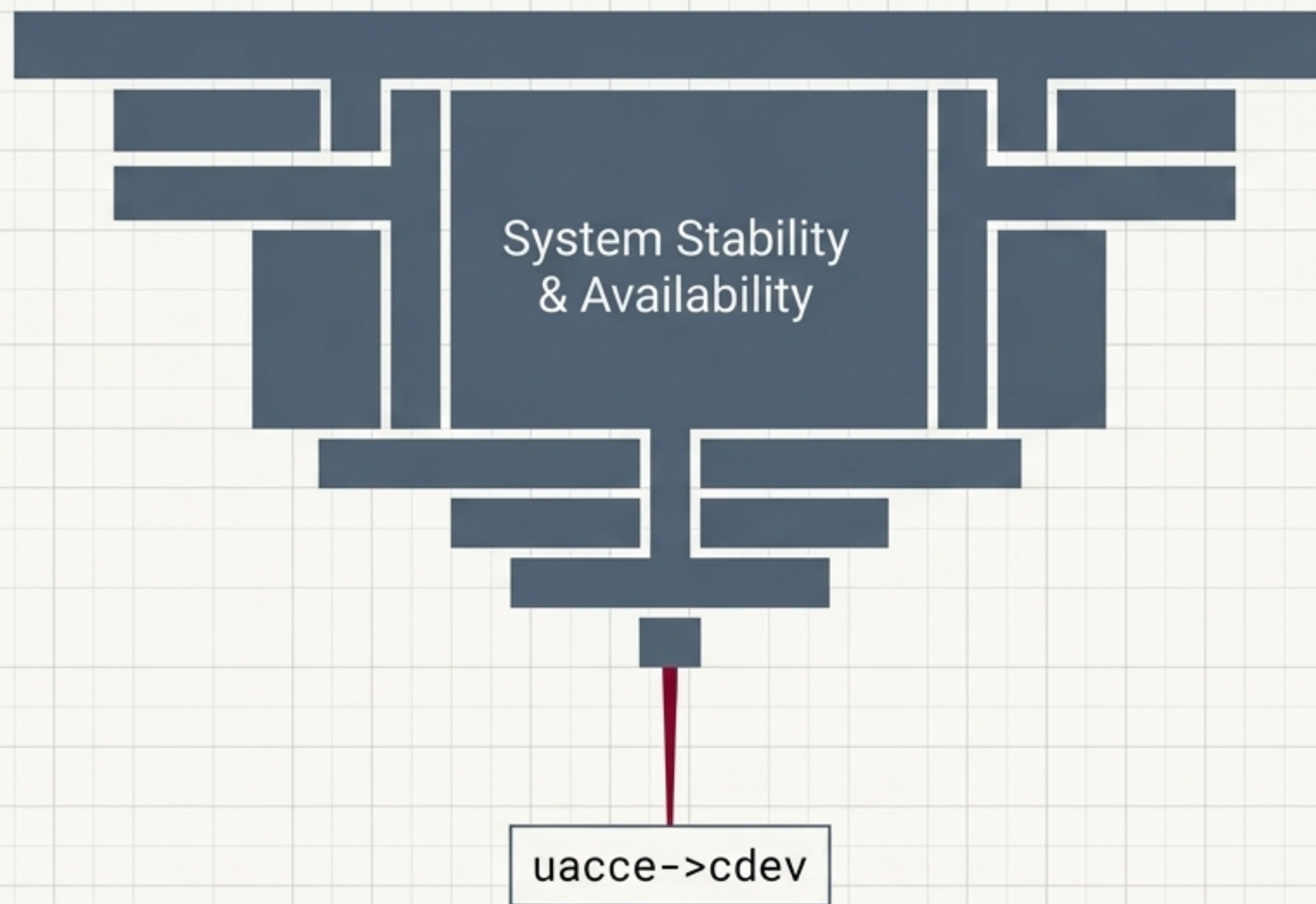
The Logic Fork: State Management Failure



Exploit Mechanics: The Phantom Pointer



The Fragility of Privilege



Kernel drivers execute with the absolute highest system privileges (Ring 0). Consequently, a microscopic logical oversight in an edge-case error handler bypasses all user-space protections.

A simple failure to update a status flag after a routine initialization error becomes a fatal weapon, transforming a minor device registration hiccup into a permanent infrastructure outage.

Tactical Mapping (MITRE ATT&CK)

T1499

**Endpoint Denial
of Service**

Resulting from the
induced kernel
freeze.

T0814

**System Resource
Exhaustion
Exhaustion**

Resulting from
duplicate memory
operations.

T1068

**Kernel-Level
Abuse
Opportunity**

Exploitation of
high-privilege driver
logic.

Telemetry and Indicators of Compromise

```
> journalctl -k | grep -i uacce  
> dmesg | tail -n 50
```

(Monitoring for unexpected device registration errors immediately preceding system unresponsiveness)

Target Conditions to Monitor

- Kernel Driver Initialization Failures
- Unexpected UACCE Registration Errors
- Warning messages emitted just prior to Kernel Hang Events

The Vendor Fix: Pointer Sanitization

Step 1: Failure Detected in Registration.

`uacce->cdev = NULL`

Step 3: Cleanup Path Triggered
(`cdev_device_del()`)

Step 4: Sees NULL. Safely aborts.

Linux developers introduced strict return-value validation. By explicitly clearing the `cdev` pointer upon failure, the secondary cleanup routine is safely bypassed.

Remediation and Operational Controls

Fixed Kernel Versions	
5.10.249+	5.15.199+
6.1.162+	6.6.122+
6.12.68+	6.18.8+
6.19+	

Apply Kernel Updates

Deploy the patched versions across all exposed infrastructure.

Restrict Drivers

Disable or restrict access to unnecessary hardware accelerator drivers via kernel module blacklisting if UACCE is not required.

Audit Telemetry

Ensure kernel logs (dmesg, journalctl -k) are actively forwarded to SIEM for continuous monitoring of initialization failures.